

1 重要公式

1.1 三角函数

1.1.1 诱导公式

奇变偶不变，符号看象限（等式左边的角所在象限的正负）

$$1. \sin\left(\frac{\pi}{2} \pm \alpha\right) = \cos \alpha$$

$$2. \sin(\pi \pm \alpha) = \mp \sin \alpha$$

$$3. \cos\left(\frac{\pi}{2} \pm \alpha\right) = \mp \sin \alpha$$

$$4. \cos(\pi \pm \alpha) = -\cos \alpha$$

1.1.2 和差公式

$$1. \sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$2. \cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$3. \tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

$$\bullet \tan\left(\frac{\pi}{4} - \alpha\right) = \frac{1 - \tan \alpha}{1 + \tan \alpha}$$

$$4. \cot(\alpha \pm \beta) = \frac{\cot \alpha \cot \beta \mp 1}{\cot \alpha \pm \cot \beta}$$

1.1.3 倍角公式

$$1. \sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$2. \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 1 - 2 \sin^2 \alpha = 2 \cos^2 \alpha - 1$$

$$3. \sin 3\alpha = -4 \sin^3 \alpha + 3 \sin \alpha$$

$$4. \cos 3\alpha = 4 \cos^3 \alpha - 3 \cos \alpha$$

1.1.4 半角公式

$$1. \sin^2 \frac{\alpha}{2} = \frac{1}{2}(1 - \cos \alpha)$$

$$2. \cos^2 \frac{\alpha}{2} = \frac{1}{2}(1 + \cos \alpha)$$

1.2 一元二次方程

一元二次方程 $ax^2 + bx + c = 0 (a \neq 0)$

1. 判别式 $\Delta = b^2 - 4ac$

- $\Delta \geq 0$, 方程有两个实根 $x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

- $\Delta < 0$, 方程有两个共轭的复根 $x_{1,2} = \frac{-b \pm \sqrt{4ac - b^2}i}{2a}$

2. 根与系数的关系 (韦达定理)

- $x_1 + x_2 = -\frac{b}{a}$

- $x_1 x_2 = \frac{c}{a}$

1.3 因式分解

1. $(a + b)^2 = a^2 + 2ab + b^2$

2. $(a - b)^2 = a^2 - 2ab + b^2$

3. $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

4. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

5. $a^2 - b^2 = (a + b)(a - b)$

6. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

7. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

8. 二项式定理

$$\begin{aligned}(a + b)^n &= \sum_{k=0}^n C_n^k a^{n-k} b^k \\&= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 \\&\quad + \cdots + \frac{n(n-1) \cdots (n-k+1)}{k!}a^{n-k}b^k \\&\quad + \cdots + nab^{n-1} + b^n\end{aligned}$$

1.4 阶乘与双阶乘

1. $n! = 1 \cdot 2 \cdot 3 \cdots n$, 规定 $0! = 1$

2. $(2n)!! = 2 \cdot 4 \cdot 6 \cdots 2n = 2^n \cdot n!$

3. $(2n-1)!! = 1 \cdot 3 \cdot 5 \cdots (2n-1)$